

BACKLOG BUDDY ? STUDY MATERIAL

Theory of Computation ? Unit-wise Lecture Notes

Subject Code: CS504 Semester: 5

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Condensed notes covering all five units: Finite Automata, Regular Expressions and Languages, Context Free Grammars, Pushdown Automata, Turing Machines and Complexity.

Unit 1: Finite Automata

- Definition, key concepts and significance of Finite Automata
- Important diagrams / derivations / algorithms for Finite Automata
- Solved examples and previous-year questions on Finite Automata

Unit 2: Regular Expressions and Languages

- Definition, key concepts and significance of Regular Expressions and Languages
- Important diagrams / derivations / algorithms for Regular Expressions and Languages
- Solved examples and previous-year questions on Regular Expressions and Languages

Unit 3: Context Free Grammars

- Definition, key concepts and significance of Context Free Grammars
- Important diagrams / derivations / algorithms for Context Free Grammars
- Solved examples and previous-year questions on Context Free Grammars

Unit 4: Pushdown Automata

- Definition, key concepts and significance of Pushdown Automata
- Important diagrams / derivations / algorithms for Pushdown Automata
- Solved examples and previous-year questions on Pushdown Automata

Unit 5: Turing Machines and Complexity

- Definition, key concepts and significance of Turing Machines and Complexity
- Important diagrams / derivations / algorithms for Turing Machines and Complexity
- Solved examples and previous-year questions on Turing Machines and Complexity

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.